Autonomous Mobility

CES 2025 stories: In-cabin monitoring technology innovations

By **Eva Sharma** - Jan 16 2025

The in-cabin monitoring technology sector is poised for continued growth, driven by technological advancements, regulatory mandates, and increasing consumer awareness of vehicle safety. This technology sector has recently witnessed significant advancements, particularly highlighted during CES 2025. These developments underscore the industry's commitment to enhancing vehicle safety, comfort, and user experience.

Here are some of the notable CES 2025 stories from the leading players-

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1. Magna
Magna International was 2025 honoree of Innovation Awards in Vehicle Tech & Advanced Mobility unveiling its ICS 21, a cutting-edge radar system designed for the interior cabin space. Its primary purpose is to detect occupants within a vehicle,

Meanwhile, LG Innotek showcased its RGB-IR In-Cabin Camera Module for AD and ADAS applications. Featuring a 5-megapixel RGB (red, green, blue) and IR (infrared) sensor, this module can monitor the driver's condition in real-time, helping to combat drowsiness.

3. Smart Eye
Smart Eye unveiled groundbreaking technologies that revolutionize in-cabin safety and interaction. Highlights include Sheila, the empathetic AI Co-Driver; Digital Mirrors for enhanced blind-spot visibility; a cutting-edge Synthetic Data Generation Tool; and industry-leading Driver Monitoring Systems and Interior Sensing AI solutions. Some of its key technology partners include Green Hills Software, STMicroelectronics, and Sonatus.

4. NOVELIC
NOVELIC demonstrated ACAM – A Single-Module In-Cabin Monitoring Solution to its prospect customers at CES2025. Its perception software stack combined with a radar sensor enables in-cabin monitoring with full interior coverage. ACAM does not require a line of sight to detect people and does not invade passenger privacy.

5. Emotion 3D
Emotion 3D, an in-cabin analysis software provider and BHTC, a leader in automotive display and control systems has collaborated, to showcase an innovative joint system of display and driver monitoring technologies at CES 2025. ART also joined hands with emotion3D to showcase a joint system highlighting the seamless integration of in-cabin analysis technology with advanced human-machine interaction. This innovative system leverages emotion3D's CABIN EYE software stack to analyze the driver's & passenger's gaze to trigger intelligent interactions with ART's HMI solution.

6. APTIV
Aptiv showcased its next-generation in-cabin experiences which is AI/ML-driven system that can utilize a wide range of in-cabin cameras - including those mounted in the rear-view mirror - and other advanced sensing technologies to provide a robust understanding of the environment inside the vehicle and enable more intelligent solutions.

7. Harman
HARMAN displayed the Ready Care in-cabin monitoring product, which uses AI and neuroscience to monitor driver behavior. The product aims to improve safety and comfort by reducing stress and enhancing focus.

8. MulticoreWare
MulticoreWare and Cipia presented an advanced in-cabin monitoring system at CES 2025, combining 60GHz radar and infrared camera technologies. This innovative fusion enables accurate tracking of driver and occupant vital signs, even in challenging conditions, to enhance real-time monitoring and proactive safety measures.

9. Gentex Corporation
Gentex showcased its mirror-integrated Driver Monitoring System (DMS) in a vehicle demonstrator at CES. The system monitors driver head position, eye gaze, and other metrics to assess distraction, drowsiness, sudden illness, and the readiness to resume manual control in semi-autonomous vehicles. It can also be expanded to incorporate 2D and structured-light-based 3D cabin monitoring, enabling the detection of passengers, behaviors, objects, and even signs of life.

9. Neonode
Neonode showcased its innovative Driver and In-cabin Monitoring System (DMS) at CES 2025. Its technology is aimed at revolutionizing the in-vehicle experience, making it safer and more intuitive for drivers and passengers alike. Our DMS is built on our proprietary MultiSensing® technology, making it possible to introduce any feature at any time with superior accuracy.

10. Seeing Machine
Seeing Machines, at CES2025, demonstrated its immersive technology in a purpose-built test car, highlighting the latest software and algorithm advancements for its FOVIO driver and occupant monitoring system (DMS/OMS) technology. Its technology was found integrated into various tier 1 customers who are also exhibiting at the CES 2025. For instance, the company's Guardian Generation 3 software, which features into Ambarella's solutions. Also, Seeing Machines is working with QNX to integrate its interior sensing technology with QNX Cabin to monitor driver drowsiness and distraction.

Comparative Analysis of In-Cabin Sensing Technologies Displayed by Key Players at CES 2025

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			Interior Monitoring System (DIMS)	emotional state; alerts for driver alertness and well-being
LG Innotek	RGB-IR In-Cabin Camera Module	5-megapixel RGB and IR sensors	DMS	Detects driver's condition; real-time monitoring; enhances safety by addressing drowsiness
Smart Eye	Sheila: Empathetic AI Co-Driver	Advanced sensing technologies, generative AI	DMS	Provides context-aware, emotionally intelligent interactions to enhance safety and comfort
Smart Eye	3D Depth Sensing	Depth sensing technology	In-Cabin	Enhance occupant safety
Novelic	In-Cabin Monitoring Solutions	Radar-based sensors	In-Cabin	Monitors occupant presence and behavior
APTIV	In-Cabin Sensing Platform	Cameras, radar, and ultrasonic sensors	In-Cabin	Enhances occupant safety and comfort through monitoring
Harman	In-Cabin Monitoring System	Cameras and sensors	In-Cabin	Detects drowsiness, distraction; personalized cabin experiences; integrates with infotainment systems
Cipia & MulticoreWare	Next-Gen In-Cabin Monitoring System	60GHz radar and infrared cameras	In-Cabin	Tracks driver and occupant vital signs for safety interventions
Gentex Corporation	Mirror-Integrated DMS	Cameras, possibly infrared sensors	DMS	Tracks driver metrics to assess distraction, drowsiness, and readiness for manual control
Neonode	In-Cabin Sensing Solutions	Optical Sensors	In-Cabin	Gesture recognition; touchless control and monitoring within the vehicle cabin
Seeing Machine	Driver Monitoring System	Cameras and AI algorithms	DMS	Detects distraction, drowsiness; real-time alerts; integrates with vehicle safety systems

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